

VEGREVILLE, AB, Canada

WMO: 714580

Lat: 53.5053N

Lon: 112.0978W

Elev: 639

StdP: 93.88

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-30.5	-38.3	0.1	-33.7	-34.8	0.2	-30.5	12.2	-3.3	10.8	-5.4	2.4	170	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.5	17.2	26.7	16.6	25.0	15.9	19.2	25.6	18.1	24.2	17.1	22.8	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.2	21.5	15.9	12.2	20.8	14.8	11.4	19.4	57.5	25.7	53.8	24.3	50.4	22.7	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.3	8.1	-38.2	31.6	4.4	1.6	-41.3	32.8	-43.9	33.7	-46.3	34.7	-49.5	35.9		
(5)			-38.4	21.4	4.2	1.3	-41.4	22.4	-43.9	23.2	-46.3	23.9	-49.3	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-13.3	-11.8	-6.0	3.4	10.3	14.5	16.9	15.5	10.6	3.5	-5.7	-12.0	(6)
(7)		DBStd	12.55	8.97	7.69	7.90	5.44	4.49	3.16	2.85	3.24	4.27	4.69	7.03	8.14	(7)
(8)		HDD10.0	3493	723	611	496	203	51	4	0	2	43	206	471	683	(8)
(9)		HDD18.3	5910	981	844	754	447	249	121	62	96	232	461	721	942	(9)
(10)		CDD10.0	659	0	0	0	6	61	139	214	174	61	4	0	0	(10)
(11)		CDD18.3	35	0	0	0	0	2	5	18	9	1	0	0	0	(11)
(12)		CDH23.3	776	0	0	0	3	71	112	280	227	81	2	0	0	(12)
(13)		CDH26.7	163	0	0	0	0	13	19	66	46	19	0	0	0	(13)
(14)	Wind	WSAvg	3.7	3.6	3.5	3.8	4.2	4.3	3.9	3.5	3.3	3.6	3.9	3.5	3.5	(14)
(15)	Precipitation	PrecAvg	413	19	11	17	30	44	70	81	54	33	19	19	16	(15)
(16)		PrecMax	574	69	23	37	65	118	128	139	103	82	44	77	39	(16)
(17)		PrecMin	221	3	1	6	2	7	21	30	11	4	1	1	3	(17)
(18)		PrecStd	73	15	6	8	16	22	28	27	19	21	11	16	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	5.2	11.3	22.8	28.5	28.9	31.2	30.3	29.3	21.4	11.9	6.7	(19)
(20)			MCWB	2.1	2.0	5.8	11.1	14.1	16.8	19.3	18.2	15.8	12.0	6.6	2.3	(20)
(21)		2%	DB	3.7	3.1	7.6	19.0	25.2	26.3	28.3	27.9	25.6	17.1	7.6	4.1	(21)
(22)			MCWB	1.3	0.7	3.5	9.3	12.8	15.3	18.6	17.7	14.6	9.7	3.8	1.3	(22)
(23)		5%	DB	1.8	1.3	5.0	15.6	22.6	24.2	26.4	25.9	22.7	14.2	5.2	1.8	(23)
(24)			MCWB	0.0	-0.5	1.8	7.6	11.9	14.7	17.7	17.0	13.6	7.9	2.2	-0.4	(24)
(25)		10%	DB	-0.7	-0.7	3.3	12.7	20.1	22.3	24.5	24.0	19.7	11.4	3.1	-0.5	(25)
(26)			MCWB	-2.3	-2.4	0.9	6.2	10.6	13.9	16.8	16.3	12.2	6.4	1.0	-2.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.3	6.3	11.7	15.4	18.6	21.2	20.6	16.6	12.4	6.6	2.6	(27)
(28)			MCDWB	4.9	4.5	10.6	21.2	24.3	25.9	27.7	26.4	25.4	20.3	11.1	5.5	(28)
(29)		2%	WB	1.4	0.9	3.8	9.7	13.7	17.0	19.7	19.0	15.2	10.2	4.2	1.2	(29)
(30)			MCDWB	3.5	2.7	7.0	17.5	22.6	23.1	26.2	25.3	24.0	15.4	7.0	3.6	(30)
(31)		5%	WB	0.0	-0.6	2.2	8.0	12.6	15.9	18.6	17.9	14.0	8.6	2.5	-0.4	(31)
(32)			MCDWB	2.0	1.2	4.7	14.7	20.7	21.6	24.6	24.0	21.8	13.2	4.7	1.6	(32)
(33)		10%	WB	-2.3	-2.4	1.1	6.5	11.4	14.8	17.6	16.8	12.7	6.9	1.1	-2.4	(33)
(34)			MCDWB	-0.7	-0.8	3.2	11.8	18.9	20.4	23.0	22.6	18.8	10.9	2.9	-0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	10.9	10.0	11.8	13.7	12.2	12.3	13.5	13.9	11.7	9.6	10.0	(35)
(36)			MCDBR	11.5	12.4	11.1	17.7	18.1	16.3	16.3	17.6	20.1	17.0	11.9	11.6	(36)
(37)			MCWBR	9.7	10.4	7.7	9.6	8.2	7.4	7.8	8.6	10.0	10.1	8.4	9.6	(37)
(38)		5% WB	MCDBR	11.5	12.0	10.7	16.4	16.1	13.0	14.5	15.3	18.3	15.8	11.1	11.6	(38)
(39)			MCWBR	9.7	10.2	7.7	9.4	7.6	6.5	7.5	8.1	9.4	9.8	8.1	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.266	0.307	0.333	0.349	0.365	0.362	0.373	0.311	0.278	0.252	0.239	(40)	
(41)		taud	2.300	2.332	2.314	2.359	2.366	2.349	2.384	2.324	2.481	2.544	2.440	2.298	(41)	
(42)		Ebn at Noon	736	852	879	899	899	884	880	846	871	838	750	684	(42)	
(43)		Edh at Noon	57	77	98	108	113	117	111	111	83	62	50	47	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.98	2.03	3.56	4.68	5.78	6.00	6.17	5.06	3.52	2.03	0.97	0.72	(44)	
(45)		RadStd	0.11	0.15	0.28	0.43	0.41	0.41	0.31	0.36	0.37	0.26	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page